GE Energy

GEK-SA-1053

Distributed Control System (DCS)
Network Switch
Instruction Guide

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Overview

This document provides instructions for configuring network switches in a GE Energy Distributed Control System (DCS) network. It includes examples for the three main switches:

- Enterasys™ N-Series
- Allied Telesyn AT8624
- Allied Telesyn AT8516

Physical connection to the switch is through a serial cable.

To connect to a particular switch, you must know the IP address(es), username and password pairs, switch location, and the operations to be performed. Levels of access include view-only, management, and administration.

Once communication is complete, log off the switch and (optionally) unplug the physical connection (when using a serial port). If a configuration file was retrieved from the switch, rename and save it. If a change was made to the switch configuration, make a notation of the change for the appropriate change management system or authority.

Network protocols used to connect to, and interact with, DCS network switches include:

Ping is a low-level network protocol used for sending small messages between machines.

Trivial File Transfer Protocol (TFTP) is a standard protocol for transferring files over a local network. It does not provide user authentication.

Hypertext Transfer Protocol (HTTP) allows you to graphically connect to, and interact with, a device using a web browser.

Serial allows you to directly connect to a device. No network is required, but a serial port is required on both ends of the connection. Most network switches are equipped with a serial port.

Telnet is a network protocol that allows you to remotely connect to a device, through a network, using a command prompt to interact with it.

Secure Shell (SSH) is similar to Telnet but is performed over an encrypted connection and uses a command prompt interface.

Network		**Serial**	
Software	PuTTY	PuTTY	
	HyperTerminal	HyperTerminal	
	Internet Explorer		
Protocols	Telnet, SSH, HTTP	Speed	9600, 8N1
Address	IP address, client switch	COM 1, COM 2	
Physical	RJ-45	DB9	
	Ethernet cable	Serial port	
	Network cable	Serial cable	

Applications

Several applications are available for connecting to network switches.

HyperTerminal

HyperTerminal is provided with most versions of Windows® prior to Windows 7. It allows you to connect to devices through both a serial port and Telnet. It can be used to access the command prompt of Allied Telesyn network switches through serial ports or Telnet; however, Enterasys switches are accessed through serial ports only.

PuTTY

PuTTY, a terminal emulator application that can operate as a client for both SSH and Telnet, is available for multiple operating systems. It runs on all recent versions of Windows, and can connect with a multitude of protocols. The command prompt of Allied Telesyn network switches can be accessed through serial ports or Telnet. Enterasys switches are accessed through serial ports or SSH. The PuTTY application is available on the GE Energy Application Depot for GE Energy employees, or from the Web. PuTTY is installed by copying the binary program to a convenient folder. It can be run from the Graphical User Interface (GUI) or the command prompt. The command prompt can be used to specify all connection parameters.

Note A Web browser (such as Internet Explorer) is required to connect to an HTTP-enabled network device.

Network Connections

The Network Topology drawing (4108 Network Drawing) provides IP addresses, as well as other information. The host name of the switch being configured is used to verify the switch, either by pinging it or connecting to it remotely. Local machine can be a temporary IP address. If so, it must be a valid IP address for that particular subnet that is not used by any other system on the network. If a serial connection is being used, this is not required. From the command prompt, type *ipconfig* to find the system IP address(es) and subnet masks on the local computer. Or from the Control Panel, select Network Connection. The TFTP server IP address is used for transferring the configuration file through TFTP.

A computer without a DB9 serial port will require a USB-to-serial adapter to connect to a serial port on the switch.

Interface Connections

Switch Types	Serial Connection Types	Network Protocol
Allied Telesyn		
AT8624 (9600 baud, 8N1)	DB9	Telnet, HTTP
AT8516 (9600 baud, 8N1)	DB9	Telnet, HTTP
Enterasys		
N3, N5, N7 (9600 baud, 8N1)	RJ-45	SSH

Adapters are typically shipped with the devices.

The Allied Telesyn switches contain DB9 serial ports. The Enterasys switches contain a serial port that typically uses a copper-based RJ-45 connection. A port adapter is required to attach this type of connector to a DB9 serial port found on most terminals. If an adapter is not available, the alternative is to use a network protocol to connect to these switches. The internal pins of adapters are wired differently. Although some adapters look identical, they cannot all be used to connect to network switches. For example, the RJ-45 to DB9 converters shipped with UCSA controllers do not connect to Enterasys switches. The physical connection for a serial port and the protocols vary.

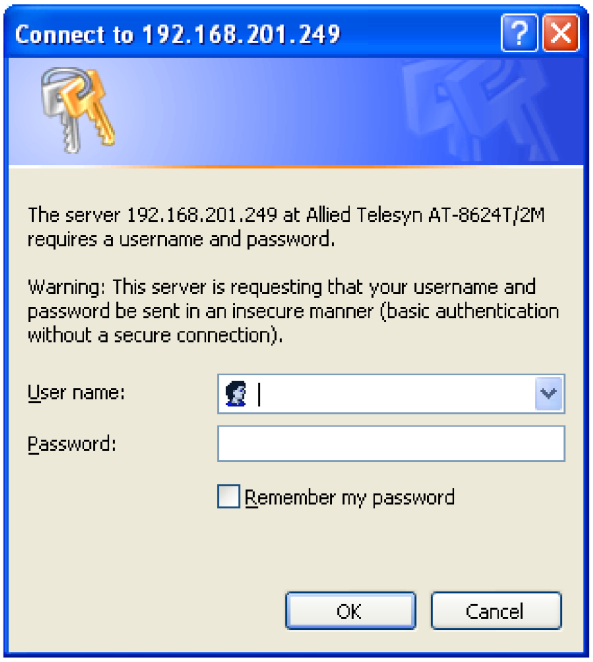
Since the GE Energy DCS network uses Virtual Local Area Network (VLAN) tagging to differentiate between Unit Data Highway (UDH) and Plant Data Highway (PDH) ports, it is necessary to understand which network the target device is on. It is recommended that the PDH be used for connecting to the switch. VLAN tagging on the Allied Telesyn AT8624 switches is standard.

VLAN tagging on the Enterasys switches is more complicated, since there is generally no single port that will always be tagged as a PDH-only port. If a network connection to an Enterasys is desired, it may be easier to physically connect to one of the other switches on the PDH (that is connected to the Enterasys), and remotely log on to the Enterasys through SSH.

HTTP Logon (AT8624)

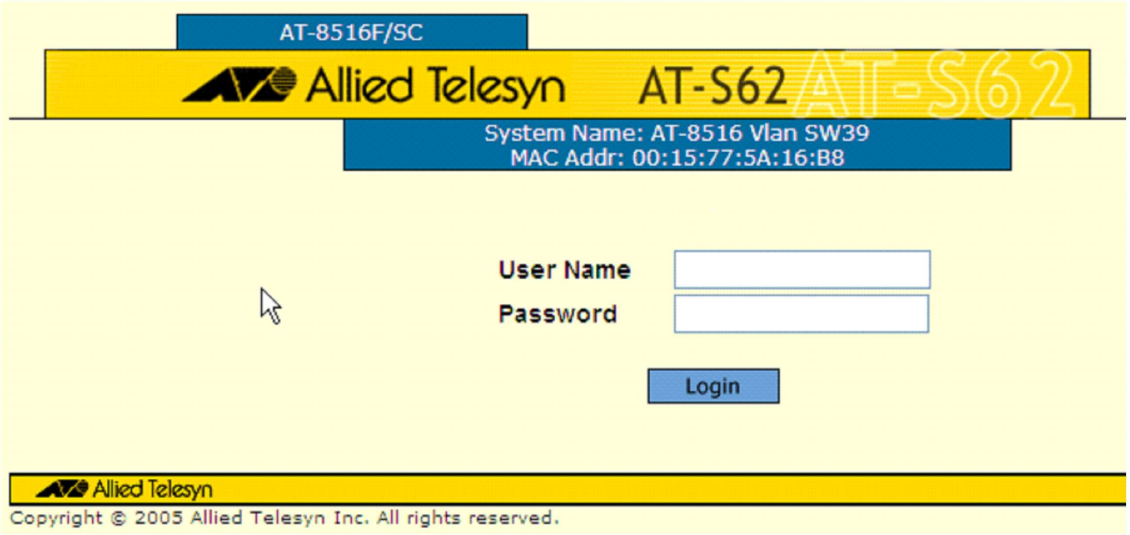
Use the default Web browser to connect to the IP address (or hostname) for the network switch. When you are connected, the initial logon dialog box displays.

Enter your **User name** and **Password** and click **OK**.



HTTP Logon (AT8516)

The AT8516 HTTP logon window is as follows:



The application used to log on through Telnet provides details on how to connect to a device.

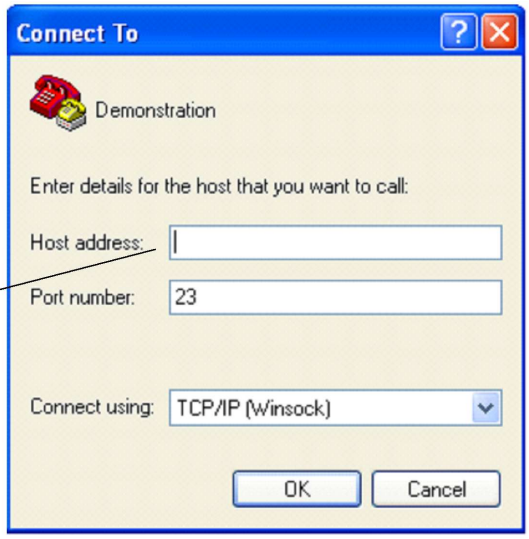
Telnet Logon with HyperTerminal

➢ **To open HyperTerminal from a Windows XP computer:** from the **Start** menu, select **Programs**, **Accessories**, **Communications**, and **HyperTerminal**.

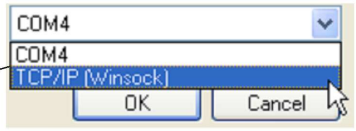
Note For a Windows 7 machine, HyperTerminal must be downloaded from the Microsoft® Web site.



Enter a name for this connection and click **OK**.



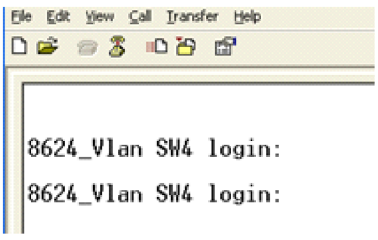
Enter the IP address of the target device and click **OK**.



HyperTerminal allows you to connect using either a serial port or a TCP/IP address.

Note If SSH is required, the PuTTY application must be used.

The following screen displays:

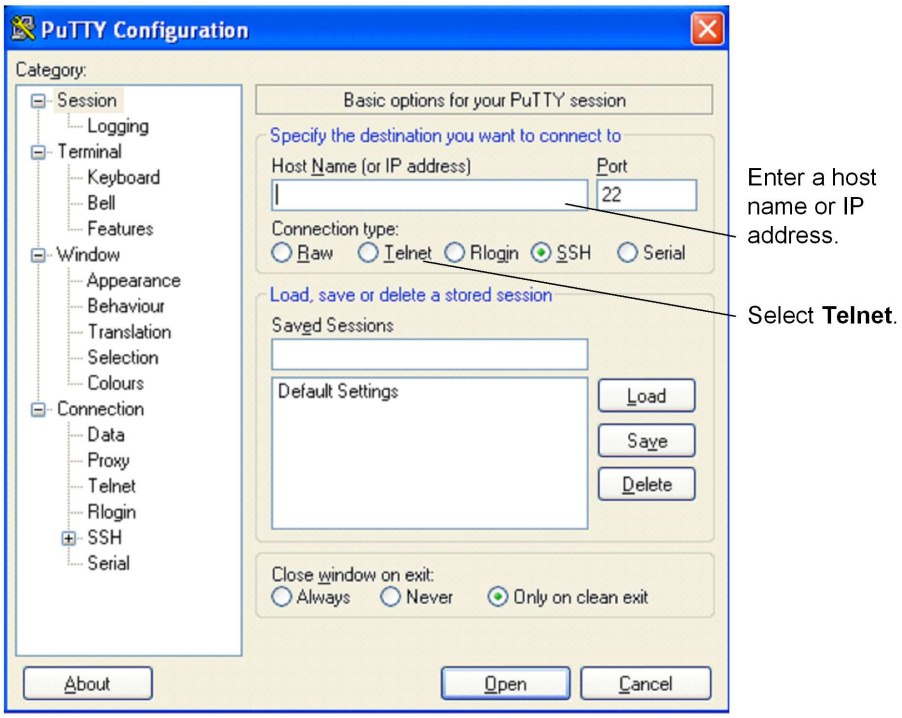


Telnet Logon with PuTTY

There are two methods to establish a Telnet session using the PuTTY application. One method is to use the command prompt:

```
C:\>
C:\>putty.exe telnet:\\sw02
```

If more configurability is required, run the PuTTY application to display the following dialog.





Attention

If SSH remains selected, and the device does not understand SSH, the connection will fail.

The following screen displays:

```
TELNET session now in ESTABLISHED state

8624_Vlan_SW01 login:
```

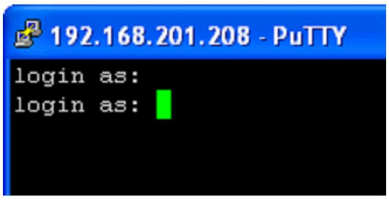
SSH Logon with PuTTY

The PuTTY application must be used for SSH connections. For a first-time network connection, the following message displays:

Click **Yes**.



Note Any new IP addresses will generate this message again.





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